

Site-Selective Cr³⁺ Occupancy and Broadband Near-Infrared Emission in Garnet-Type Phosphors

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Abstract

Cr³⁺-activated garnet phosphors are promising broadband near-infrared (NIR) emitters for compact light sources. However, the relationship between site-selective Cr³⁺ occupancy and the resulting emission profile remains debated. Here we report a series of garnet-type phosphors in which the Cr³⁺ concentration is varied systematically, and we correlate the occupancy of octahedral versus tetrahedral sites with the observed redshift of the emission maximum.

摘要：本文系统研究了石榴石型荧光粉中 Cr³⁺ 的占位行为与近红外发射特性之间的关系。结果表明，随着掺杂浓度的提高，发射峰位发生明显红移，且发光寿命随之缩短，说明浓度猝灭效应在较高掺杂量下占据主导地位。

1. Introduction

Broadband near-infrared phosphors have attracted considerable attention because they can be combined with blue light-emitting diodes to build compact and efficient NIR sources. Among the available activators, Cr³⁺ is particularly attractive owing to its tunable emission and strong absorption in the visible region.

In garnet-type hosts, Cr³⁺ can occupy both octahedral and tetrahedral cation sites, and the crystal-field strength differs substantially between them. This difference is widely believed to be responsible for the unusually broad emission band. Nevertheless, a quantitative description of how the dopant concentration controls site preference is still lacking, which limits rational design of new compositions.

2. Experimental

2.1 Sample preparation

Powder samples were synthesized by a conventional high-temperature solid-state reaction. Stoichiometric amounts of the starting oxides and carbonates were mixed thoroughly in an agate mortar, transferred to alumina crucibles, and fired in air. The Cr³⁺ content was varied from 0.5 mol% to 8.0 mol% while keeping all other cation ratios fixed.

2.2 Characterization

Phase purity was examined by X-ray diffraction. Room-temperature photoluminescence spectra were recorded on a spectrofluorometer equipped with a continuous xenon lamp, and decay curves were collected using a pulsed laser diode. All measurements were performed under identical optical alignment to allow direct intensity comparison.

3. Results and Discussion

3.1 Phase and microstructure

All diffraction patterns could be indexed to the expected garnet structure, and no impurity phase was detected up to the highest doping level. A slight contraction of the unit cell volume was observed, which is consistent with substitution of the host cation by the smaller Cr³⁺ ion. Scanning electron microscopy revealed well-faceted particles with an average size of approximately 4 micrometers.

3.2 Luminescence properties

Under blue excitation all samples exhibited a broad emission band centred in the near-infrared region. With increasing Cr³⁺ concentration, the emission maximum shifted from 748 nm to 812 nm, while the

integrated intensity first increased and then decreased, reaching a maximum at 3.0 mol%. The decay lifetime shortened monotonically, from 128 microseconds to 41 microseconds, indicating that concentration quenching becomes dominant beyond 3.0 mol%.

These trends support a model in which Cr³⁺ preferentially occupies the octahedral site at low concentration and increasingly populates the tetrahedral site as the dopant content rises. Because the tetrahedral crystal field is weaker, emission from that site is redshifted, which explains the observed shift of the band maximum.

4. Conclusions

We have shown that the near-infrared emission of garnet-type phosphors is governed by the site distribution of Cr³⁺. A doping level of 3.0 mol% provides the best compromise between absorption strength and quenching, and the corresponding emission maximum lies at 781 nm. These findings offer a practical guideline for designing broadband NIR phosphors with tailored emission profiles.

References

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